



Properties of MLEs Transcript

Hello everyone, welcome to lesson 1 video 4. In this session, we will discuss the properties of Maximum Likelihood Estimators. These properties explain why MLE is so powerful and widely used. You have now made yourself clear with the idea of Maximum Likelihood Estimation.

In this video, we are going to understand why MLE or Maximum Likelihood Estimations are so popular. There are various different types of methods for estimating the value of the parameter given the data. But this method, why is it popular or it is more statistical advantageous method? What are those properties which makes this method more advantageous is what we are going to look into.

So, in the very first argument that I want to put forward, we need to understand that what would happen the value of the parameter when we have asymptotic experiment. So, let us say we have large samples. Maximum Likelihood Estimators have excellent property as the sample size gets large.

So, this is what we call as asymptotic. When you repeat an experiment or let us say you take huge number of sample size and that becomes tending towards a very large number closer to infinity. So, something that you cannot measure it is so large.

In those cases, the estimate itself becomes consistent. What does it mean? What does consistency mean? As the sample size increases, the MLE converges into a probability that the true value of the parameter would have. In other words, you can also think about it as with enough data MLE will arbitrarily be closer to the real answer.

When you have good amount of data, your MLE or the Maximum Likelihood Estimation would be very close to what your true values are. So, they are very consistent first thing. Second, asymptotically they behave normality or they have normal distribution.

Let's say for example, when you have large sample, the sampling distribution of MLE is approximately normal and you already know that you have Poisson binomial distribution as well. When you repeat the experiment infinite number of times and you take sample averages using central limit theorem, which we already discussed, most of the things whether their parent distribution or parent population distribution is normal or not normal, whatever it is, the sampling distribution is generally normal. And CLT is the base to majority of the things that we learn about normality in real life.

So, the second thing that we have is when you have large samples, the estimate or the sampling distribution using which you calculate the estimates, they are approximately the normal distribution. This is extremely useful as it allows us to construct confidence interval around the parameter that we have got. So, it is also very important not to identify one single parameter which is the point estimate.



It is also important to create a decision boundary plus and minus such that within which there are more chances the value of the parameter resides. So, that can be calculated or that can be well calculated when you know that your parameter or the distribution from which the parameter is obtained, it is following a normal distribution. So, it gives a lot of good mathematical advantages so that we can create confidence interval, we can estimate things more likely, you can also have things which are more deterministic than non-deterministic.

Third is the efficiency. So, MLE has the smallest possible variance among all the consistent asymptotically normal estimators. So, because it is having least variance, least possible variance, hence they are all called as efficient.

So, this is also one of the properties of estimators if you remember in the third or fourth module we discussed about various different properties of estimators what are the best properties of estimators that make them the best estimators. So, this is also one of the properties of the estimators which describes about the variance among all the estimates that we get. Because MLE has the smallest possible variance among all the estimators, hence it is very efficient.

I hope this entire logic is embedded in your head right now. Now, invariance property is also what makes MLE very convenient. Let us say if t is the MLE of a parameter θ , then for any function $g(\theta)$, the MLE of $g(\theta)$ is simply $g(t)$.

So, let us say we find MLE of the variance for a normal distribution. We have found an estimate of the variance for a normal distribution. If we want the MLE for the standard deviation, we do not need to start all over again.

We do not have to recompute it. So, the invariance property says that we just take the square root of MLE and then we will be able to get to the standard deviation. This is exactly what we use to find out the value of standard deviation if you remember.

This gets to that because we have a convenient property which is called as the convenient transformation or the invariance property. So, in summary, you have learned about a lot of good things in this entire lecture. MLE and its importance and MLE, why MLEs are also the best estimates.

The summarization about all the things that we have learned in this could be, firstly, it is based on the very intuitive principle that find the parameter given the data. In this entire module, you learned about all of those things. Second, provides a clear recipe for finding estimators.

You know how to find the estimators. You know how to use calculus and then take log likelihood, take the calculus, use calculus, find the value of the parameter which



maximises the likelihood and so on. Then, you have laid down the good excellent properties or good properties about the maximum likelihood estimation which are consistency, normality, and efficiency.

Consistency is they are all consistent. So, let us say if you take a very large sample, the parameter estimate that you will be getting via MLE would be consistent. The distribution of the parameter estimate would be almost near to normal.

The variance of the parameter estimate would be least among all the parameter estimates that you could get. Because of all of these properties, MLEs are most widely used in statistics. I hope you would use it in the next module that we are going to learn because in the next module when we start discussing about finding the parameters of the model that we are going to learn, we will understand how do we use that in understanding or finding out the parameters.

In summary, maximum likelihood estimators are consistent, normal, efficient, making the gold standard in statistical inference. Together, the intuition for maximum likelihood function gives us a wide range of things to study the data. Thank you.

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